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Domain :- AI - ML

Branch Name :- Computer Engineering

1)

df.head()

	age	sex	bmi	children	smoker	region	expenses
0	19	female	27.9	0	yes	southwest	16884.92
1	18	male	33.8	1	no	southeast	1725.55
2	28	male	33.0	3	no	southeast	4449.46
3	33	male	22.7	0	no	northwest	21984.47
4	32	male	28.9	0	no	northwest	3866.86

Next steps: [Generate code with df](#) [New interactive sheet](#)

2)

```
9] df.isnull().sum()
***
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
expenses 0
dtype: int64

10] df.duplicated().sum()
np.int64(0)

12] df = df.drop_duplicates()
```

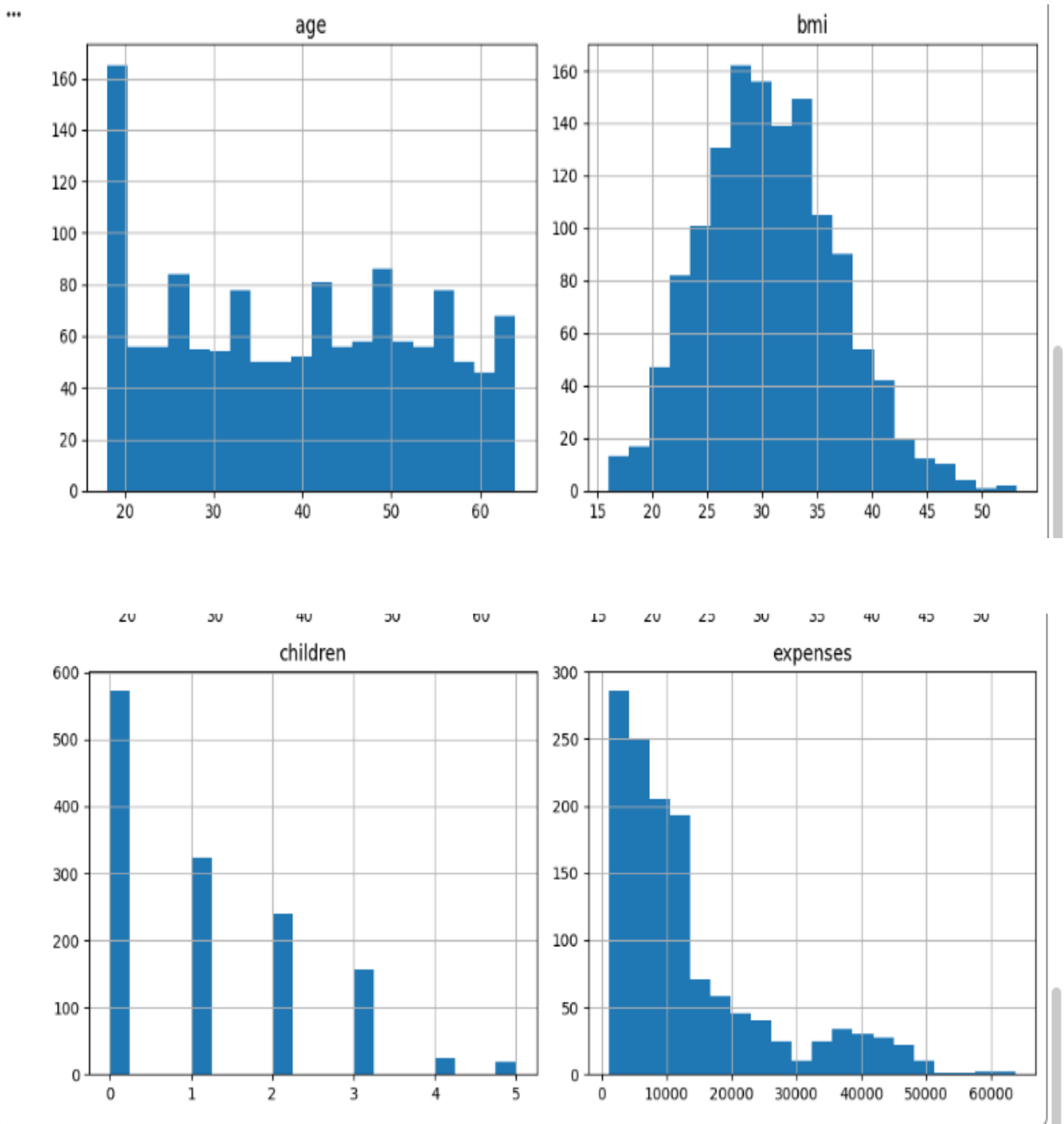
3)

```
df.shape
df.info()
df.describe()
```

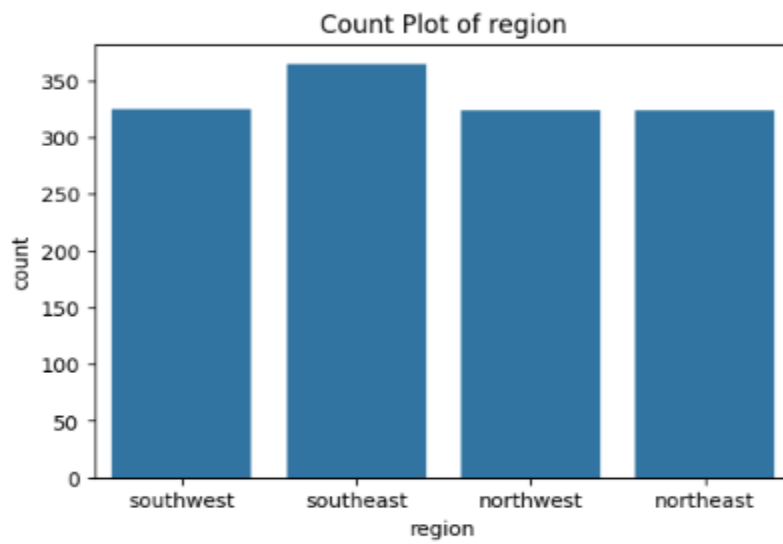
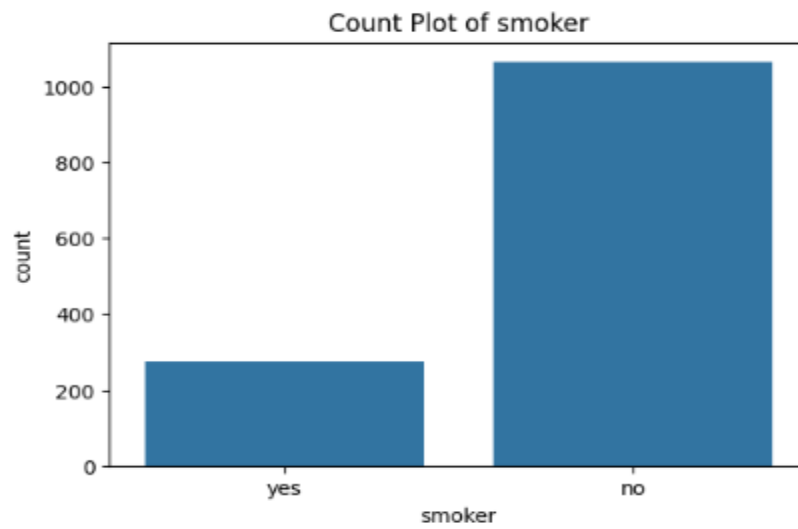
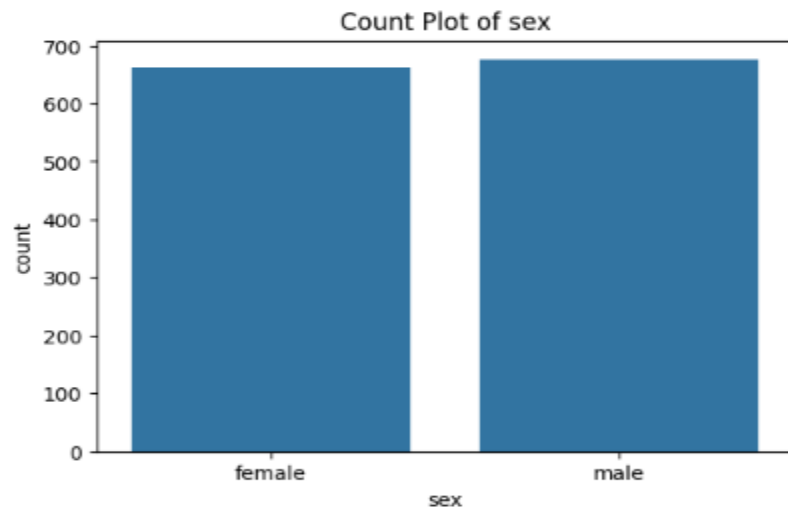
```
*** <class 'pandas.core.frame.DataFrame'>
Index: 1337 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   age         1337 non-null   int64
1   sex         1337 non-null   object
2   bmi         1337 non-null   float64
3   children    1337 non-null   int64
4   smoker      1337 non-null   object
5   region      1337 non-null   object
6   expenses    1337 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 83.6+ KB
```

	age	bmi	children	expenses
count	1337.000000	1337.000000	1337.000000	1337.000000
mean	39.222139	30.665520	1.095737	13279.121638
std	14.044333	6.100664	1.205571	12110.359657
min	18.000000	16.000000	0.000000	1121.870000
25%	27.000000	26.300000	0.000000	4746.340000
50%	39.000000	30.400000	1.000000	9386.160000
75%	51.000000	34.700000	2.000000	16657.720000
max	64.000000	53.100000	5.000000	63770.430000

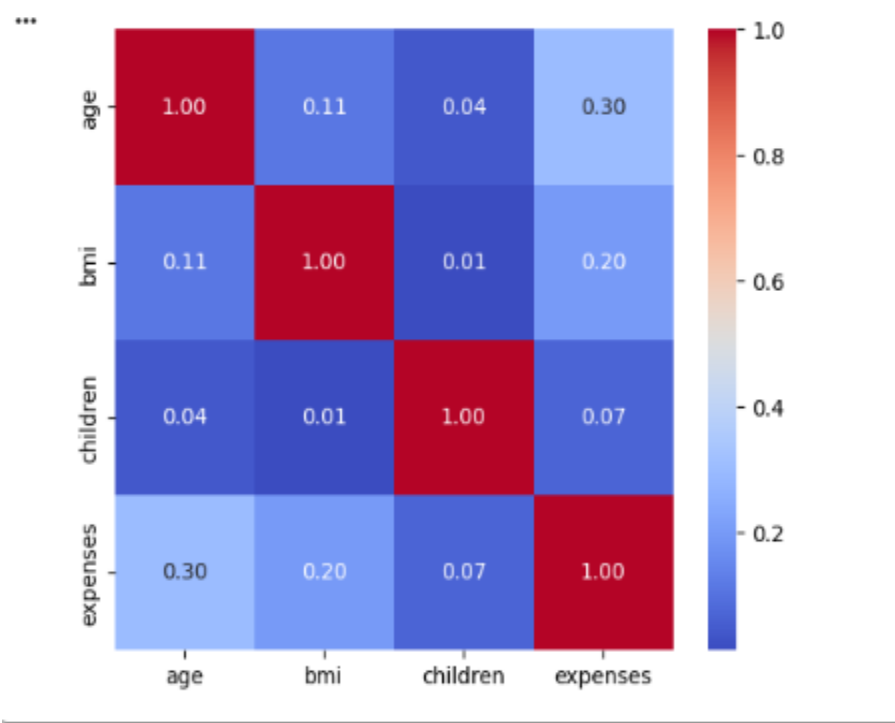
4)



5)



6)



7)

	age	sex	bmi	children	smoker	region
0	-1.440418	female	-0.453484	-0.909234	yes	southwest
1	-1.511647	male	0.513986	-0.079442	no	southeast
2	-0.799350	male	0.382803	1.580143	no	southeast
3	-0.443201	male	-1.306169	-0.909234	no	northwest
4	-0.514431	male	-0.289506	-0.909234	no	northwest
